

1.2

$$\langle f, g \rangle = \int_1^5 f(x)g(x) dx$$

a, $\text{span} \{1, x, x^2\} = \text{span} \{u_1, u_2, u_3\} = V$

$$u_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\|1\|} = \frac{1}{2} \quad \sqrt{\|1\|^2} = \sqrt{\int_1^5 1^2 dx} = \sqrt{4} = 2$$

$$u_2 = \frac{u_2 - w_1}{\|u_2 - w_1\|}, \quad w_1 = \langle u_1, u_2 \rangle u_1 = \left(\int_1^5 \frac{x}{2} dx \right) \frac{1}{2} = \left(\frac{x^2}{4} \right) \frac{1}{2} = \frac{x^2}{8}$$

$$u_2 = \frac{u_2 - w_1}{\|u_2 - w_1\|} = \frac{x - \frac{x^2}{8}}{\sqrt{\int_1^5 (x - \frac{x^2}{8})^2 dx}} = \frac{x - \frac{x^2}{8}}{\sqrt{\frac{16}{5}}} = \frac{x - \frac{x^2}{8}}{\frac{4}{\sqrt{5}}} = \frac{\sqrt{5}}{4} (x - \frac{x^2}{8})$$

$$u_3 = \frac{u_3 - w_1}{\|u_3 - w_1\|}$$

$$w_2 = \langle u_1, u_3 \rangle u_1 + \langle u_2, u_3 \rangle u_2 = \left(\int_1^5 \frac{x^2}{2} dx \right) \frac{1}{2} + \left(\int_1^5 \frac{\sqrt{5}}{4} (x - \frac{x^2}{8}) x^2 dx \right) \frac{\sqrt{5}}{4} (x - \frac{x^2}{8})$$

$$= \frac{21}{5} + \frac{\sqrt{5}}{4} \left(\frac{\sqrt{5}}{4} (x - \frac{x^2}{8}) \right) = \frac{21}{5} + \frac{\sqrt{5}}{4} (x - \frac{x^2}{8})$$

$$\frac{u_3 - w_2}{\|u_3 - w_2\|} = \frac{x^3 - (x + \frac{x^2}{8})}{\sqrt{\int_1^5 (x^3 - (x + \frac{x^2}{8}))^2 dx}} = \frac{x^3 - 6x + \frac{x^2}{8}}{\sqrt{\frac{384}{5}}} = \frac{3\sqrt{5}}{16} (x^3 - 6x + \frac{x^2}{8})$$

Dobití bázi ON bázi: $B = \left(\frac{1}{2}, \frac{\sqrt{5}}{4} (x - \frac{x^2}{8}), \frac{3\sqrt{5}}{16} (x^3 - 6x + \frac{x^2}{8}) \right)$

b, Projekce vypočítáme pomocí vektoru, můžeme totiž ON bázi:

$$w = \left(\frac{x}{2}, \sin x \right) \frac{1}{2} + \left(\frac{\sqrt{5}}{4} (x - \frac{x^2}{8}), \sin x \right) \frac{\sqrt{5}}{4} (x - \frac{x^2}{8}) + \left(\frac{3\sqrt{5}}{16} (x^3 - 6x + \frac{x^2}{8}), \sin x \right) \frac{3\sqrt{5}}{16} (x^3 - 6x + \frac{x^2}{8})$$

$$= \frac{1}{4} (\cos 1 - \cos 5) - \frac{5}{16} (x - 3) (\sin 1 - \sin 5 + 2 \cos 1 + \cos 5)$$

$$+ \frac{15}{16} \left(x^3 - 6x + \frac{x^2}{8} \right) ((\sin 1 + \sin 5) + \cos 1 - \cos 5)$$

c, $\| \sin x - w \|^2 = \| \sin x \|^2 - \| w \|^2 = \| \sin x \|^2 - \left(\left(\frac{1}{2} (\cos 1 - \cos 5) \right)^2 + \left(\frac{\sqrt{5}}{4} (\sin 1 - \sin 5 + 2 \cos 1 + \cos 5) \right)^2 \right)$

$$= \left(\frac{15}{16} (6 \sin 1 + \sin 5) + \cos 1 - \cos 5 \right)^2$$

$$\approx \| \sin x \|^2 - 2,2613 = 2,5633 - 2,2613 = 0,1016$$

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